

Event Management System

USE CASE FLOWS

UC-01: Register Account | Actor: Participant / Organizer

Goal	Register a new user account on the platform
Actor	Participant / Organizer
Main flow	1. User selects "Sign Up" and is shown a form for name, email, and password. 2. User enters name, email, and password, then submits. 3. System validates the input format (non-empty name, valid email, password meets policy). 4. System creates the account in a "pending verification" state. 5. System sends a verification email containing an activation link. 6. User opens the email and clicks the verification link. 7. System activates the account. 8. Use case ends successfully.
Alternate Flow	3. Invalid Input Trigger: At Step 3, input format fails validation. 3a. System detects an invalid email or password format. 3b. System displays a specific validation error and does not create the account. 3c. User corrects the input and resubmits (resumes at Step 2). 4. Email Already Registered Trigger: At Step 4, email already exists in the system. 4a. System detects the email already belongs to an existing account. 4b. System displays "email already registered" and does not create a duplicate account (process ends). 6. Verification Link Expired Trigger: At Step 6, activation window has lapsed. 6a. User clicks a verification link past its expiry window. 6b. System displays "link expired" and offers to resend a new verification email (resumes at Step 6).

UC-02: Login | Actor: Participant / Organizer / Admin

Goal	Authenticate into the system and access the dashboard
Actor	Participant / Organizer / Admin
Main flow	1. User enters email and password on the Login page. 2. System validates the credentials against the stored account. 3. System confirms the account is verified and active. 4. System creates a session and redirects the user to their role-specific dashboard.

	5. Use case ends successfully.
Alternate Flow	2a. Incorrect Credentials At Step 2, if credentials do not match: 2a1. System detects the email/password combination does not match. 2a2. System displays "invalid credentials" and does not create a session (resumes at Step 1). 3a. Account Not Verified At Step 3, if email verification is incomplete: 3a1. System detects the account has not completed email verification. 3a2. System blocks login and prompts the user to verify their email and offers to resend link (process ends). 3b. Account Suspended or Deleted At Step 3, if account status is restricted: 3b1. System detects the account has been suspended or deleted by an Admin. 3b2. System blocks login and displays "account suspended" (or equivalent) message (process ends). Forgot Password At Step 1, if user requests password reset: F1. User clicks "Forgot Password" and enters their registered email. F2. System sends a time-limited password-reset link to that email. F3. User opens the link, sets a new password, and the old password is invalidated. F4. User logs in with the new password (resumes Main flow at Step 1).

UC-03: Browse / Search Events | Actor: Participant

Goal	Search, filter, and view event listings
Actor	Participant
Main flow	- Participant navigates to the Events page. - System displays a list of all upcoming, non-cancelled events. - Participant enters a search term and/or selects filters (category, date, venue). - System returns events matching all applied criteria. - Participant selects an event to view full details. - Use case ends successfully.
Alternate Flow	4a. No Matching Events At Step 4, if zero records match criteria: 4a1. System finds no events matching the search/filter criteria.

	4a2. System displays "no events found" and suggests clearing filters (resumes at Step 3). 3a. Category List Unavailable At Step 3, if category catalog service fails: 3a1. The admin-managed category list fails to load. 3a2. System falls back to name/date/venue filters only and logs the error (resumes at Step 3).
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UC-04: View Seat Availability | Actor: Participant

Goal	Check remaining seat capacity for a selected event
Actor	Participant
Main flow	- Participant opens an event's details page. - System retrieves the event's capacity and current active-registration count. - System displays the number of available seats alongside the event details. - Use case ends successfully.
Alternate Flow	2a. Event Is Full At Step 2, if no seats remain: 2a1. System computes zero available seats. 2a2. System displays "Fully booked" instead of a seat count and disables direct registration and offers waitlist — see UC-06 (process ends). 2b. Stale Data on Refresh Between Step 2 and subsequent action, if seat count changes: 2b1. Seat count changes between page load and a subsequent action. 2b2. System re-fetches the latest count before allowing registration to proceed (resumes at Step 2).

UC-05: Register for Event | Actor: Participant (Includes: UC-08, UC-09 | Extended by: UC-06)

Goal	Secure a confirmed seat for a specific event
Actor	Participant
Main flow	- Participant selects an event and clicks "Register." - System confirms the Participant is logged in and displays current seat availability. - Participant confirms registration. - System checks remaining capacity (<<include>> UC-08 Check Event Capacity). - System verifies the Participant has no existing active registration for this event. - System creates a registration record (user, event, timestamp, status = "confirmed") and decrements the seat count by one, atomically. - System sends a confirmation notification (<<include>> UC-09 Send Notification). - System displays a success message with a confirmation reference.

	9. Use case ends successfully.
Alternate Flow	1a. Registration Window Closed or Event Cancelled At Step 1, if event is unavailable: 1a1. System detects the event is cancelled or past its registration deadline. 1a2. System disables the Register action and displays the reason (process ends). 4a. Event Is Full At Step 4, if capacity is exhausted: 4a1. System detects zero remaining seats. 4a2. System offers the Participant the option to join the waitlist (<<extend>> UC-06 Join Waitlist). 4a3. If accepted, control passes to UC-06; if declined, use case ends without registration (process ends). 5a. Duplicate Registration At Step 5, if Participant is already registered: 5a1. System detects an existing active registration for this Participant and event. 5a2. System displays "already registered" and cancels the new attempt (process ends). 6a. Concurrent Last-Seat Conflict At Step 6, if race condition occurs: 6a1. Multiple Participants attempt to claim the last seat simultaneously. 6a2. System's atomic check-and-decrement admits exactly one registration; all others are re-evaluated as full (resumes at Alternate Flow 4a).

UC-06: Join Waitlist | Actor: Participant (Extends: UC-05 Register for Event)

Goal	Join a waiting list when an event has reached capacity
Actor	Participant
Main flow	- Participant is offered the waitlist option after Register for Event detects a full event. - Participant accepts and confirms joining the waitlist. - System records the Participant on the event's waitlist, ordered by request time. - System notifies the Participant of their waitlist position. - Use case ends successfully; Participant status is "waitlisted."
Alternate Flow	2a. Participant Declines At Step 2, if waitlist option is refused: 2a1. Participant declines the waitlist offer. 2a2. Use case ends with no waitlist entry created (process ends). 3a. Already Waitlisted At Step 3, if duplicate entry detected: 3a1. System detects the Participant is already on this event's waitlist. 3a2. System displays "already on waitlist" and does not create a duplicate entry (process ends). Waitlist Promotion

	Triggered externally by UC-07 Cancel Registration: W1. A registered Participant cancels their registration for the event. W2. System automatically promotes the first waitlisted Participant to "confirmed" status. W3. System sends a promotion notification to that Participant (<<include>> UC-09). W4. If the promoted Participant does not act within a configured window (if applicable), system may offer the seat to the next waitlisted Participant (process ends).
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UC-07: Cancel Registration | Actor: Participant (Includes: UC-09 Send Notification)

Goal	Cancel an active event registration and release the reserved seat
Actor	Participant
Main flow	- Participant opens "My Registrations" and selects an active registration. - Participant clicks "Cancel Registration." - System asks for confirmation. - Participant confirms. - System updates the registration status to "cancelled" and increments the event's available seat count by one. - System checks the event's waitlist and, if non-empty, promotes the next waitlisted Participant (see UC-06 Waitlist Promotion). - System sends a cancellation confirmation to the Participant (<<include>> UC-09 Send Notification). - Use case ends successfully.
Alternate Flow	4a. Participant Aborts Cancellation At Step 4, if confirmation is rejected: 4a1. Participant declines the confirmation prompt. 4a2. Use case ends; registration remains active/unchanged (process ends). 5a. Event Already Occurred or Already Cancelled At Step 5, if status makes cancellation invalid: 5a1. System detects the event's date has passed, or the event itself was already cancelled by the Organizer. 5a2. System blocks self-cancellation (or treats it as a no-op) and displays an explanatory message (process ends).

UC-08: Check Event Capacity | Actor: Internal (System)

Goal	Evaluate remaining seat capacity for downstream booking operations
Actor	Internal (invoked by Register for Event)
Main flow	Requesting use case (Register for Event) invokes this check with the target event ID.

	2. System retrieves the event's capacity and current confirmed-registration count. 3. System returns "available" if confirmed registrations are less than capacity, otherwise "full." 4. Use case ends successfully, returning control to the calling use case.
Alternate Flow	2a. Event Not Found or Inactive At Step 2, if target ID is invalid: 2a1. System cannot locate an active event for the given ID (deleted/cancelled). 2a2. System returns "unavailable" and the calling use case handles it as a registration failure (process ends).

UC-09: Send Notification | Actor: Notification Service (External / System)

Goal	Deliver event and account-related email notifications
Actor	Notification Service (external/system actor)
Main flow	1. A calling use case requests a notification with a template type and recipient. 2. System composes the notification content from the relevant template and event/user data. 3. System dispatches the notification via email to the recipient. 4. System logs the notification as sent. 5. Use case ends successfully.
Alternate Flow	1a. Invalid Recipient At Step 1, if email address format is invalid: 1a1. The recipient's email address is missing or malformed. 1a2. System logs the error and does not attempt delivery (process ends). 3a. Delivery Failure At Step 3, if mail service fails: 3a1. The email service is unreachable or returns an error. 3a2. System queues the notification for retry and logs the failure. 3a3. The calling use case still completes; notification delivery is not a blocking dependency (process ends).

UC-10: Create Event | Actor: Organizer

Goal	Publish a new event to the platform catalog
Actor	Organizer
Main flow	1. Organizer selects "Create Event" and is shown the event form. 2. Organizer enters name, description, date, time, venue, category, and capacity. 3. Organizer submits the form. 4. System validates all required fields and that capacity is a positive integer. 5. System creates the event and lists it as upcoming.

	6. Use case ends successfully.
Alternate Flow	2a. Category Not Yet Defined At Step 2, if category is missing: 2a1. Organizer needs a category that does not exist in the admin-managed list. 2a2. Organizer selects the closest existing category or requests the Admin add a new one (see UC-14 Manage Event Categories, resumes at Step 2). 4a. Missing or Invalid Field At Step 4, if form validation fails: 4a1. System detects a missing required field or invalid value (e.g., capacity <= 0, date in the past). 4a2. System displays a field-specific error and does not create the event (resumes at Step 2).

UC-11: Update Event | Actor: Organizer

Goal	Modify existing event details and inform confirmed participants
Actor	Organizer
Main flow	1. Organizer opens an event they own and selects "Edit." 2. Organizer modifies one or more fields (e.g., venue, time, capacity). 3. Organizer saves the changes. 4. System validates the updated fields. 5. System persists the changes and updates the event listing immediately. 6. System notifies all currently registered participants of the update (<<include>> UC-09 Send Notification). 7. Use case ends successfully.
Alternate Flow	1a. Not the Owning Organizer At Step 1, if user lacks permission: 1a1. A user who does not own the event attempts to edit it. 1a2. System denies access with an authorization error (process ends). 4a. Invalid Update At Step 4, if parameters break business logic: 4a1. System detects an invalid value (e.g., new capacity below the current number of confirmed registrations). 4a2. System rejects the update with an explanatory error; no changes are persisted (resumes at Step 2).

UC-12: Cancel Event | Actor: Organizer (Includes: UC-09 Send Notification)

Goal	Cancel an event and broadcast notices to all registrants
Actor	Organizer
Main flow	1. Organizer opens an event they own and selects "Cancel Event." 2. System asks for confirmation.

	3. Organizer confirms. 4. System marks the event status as "cancelled" and closes it to new registrations. 5. System notifies all registered and waitlisted participants of the cancellation (<<include>> UC-09 Send Notification). 6. Use case ends successfully.
Alternate Flow	2a. Organizer Aborts At Step 2, if confirmation is refused: 2a1. Organizer declines the confirmation prompt. 2a2. Use case ends; event remains active (process ends). 4a. Event Already Cancelled or Completed At Step 4, if cancellation is invalid: 4a1. System detects the event is already cancelled or has already occurred. 4a2. System blocks the action (or treats as a no-op) and displays an explanatory message (process ends).

UC-13: View Registrants | Actor: Organizer (Extended by: UC-15 Export Participant List)

Goal	Inspect the list of confirmed and waitlisted attendees for an event
Actor	Organizer
Main flow	1. Organizer opens an event they own and selects "View Registrants." 2. System retrieves all active (confirmed and waitlisted) registrations for that event. 3. System displays the registrant list with name, contact, status, and registration timestamp. 4. Use case ends successfully.
Alternate Flow	1a. Not the Owning Organizer At Step 1, if user lacks authorization: 1a1. A user who does not own the event attempts to view its registrants. 1a2. System denies access with an authorization error (process ends). 2a. No Registrants Yet At Step 2, if list is empty: 2a1. System finds zero registrations for the event. 2a2. System displays "no registrants yet" instead of an empty table (process ends successfully).

UC-14: Manage Event Categories | Actor: Admin

Goal	Add, update, or remove event categories in the system
Actor	Admin
Main flow	1. Admin opens "Manage Categories."

	2. System displays the current list of event categories. 3. Admin adds, edits, or removes a category and saves. 4. System validates the change (e.g., non-empty name, no duplicate). 5. System persists the change; the updated list is immediately available to Organizers creating/editing events. 6. Use case ends successfully.
Alternate Flow	3a. Category In Use When Deleting At Step 3, if category is associated with live events: 3a1. Admin attempts to delete a category currently assigned to one or more events. 3a2. System blocks deletion (or prompts to reassign affected events first) and explains why (resumes at Step 2). 4a. Duplicate Category Name At Step 4, if entry name collisions occur: 4a1. System detects the new/edited name already exists. 4a2. System rejects the change with a duplicate category error (resumes at Step 3).

UC-15: Export Participant List | Actor: Organizer (Extends: UC-13 View Registrants)

Goal	Export event registrant rosters to downloadable file formats
Actor	Organizer
Main flow	1. While viewing an event's registrants (UC-13), Organizer clicks "Export" and selects a format (CSV or PDF). 2. System compiles the current registrant list into the selected format. 3. System generates the file and makes it available for download. 4. Organizer downloads the file. 5. Use case ends successfully; the exported file matches the on-screen list exactly.
Alternate Flow	2a. No Registrants to Export At Step 2, if event has zero registrations: 2a1. System finds zero registrants for the event. 2a2. System still generates a valid, empty-body file (header row only) rather than failing (resumes at Step 3). 3a. Export Generation Failure At Step 3, if rendering crashes or fails: 3a1. File generation fails (e.g., server error). 3a2. System displays "export failed, please retry" and does not return a partial/corrupt file (resumes at Step 1).

UC-16: Manage User Accounts | Actor: Admin

Goal	Review, suspend, or delete user accounts across the platform
Actor	Admin

Main flow	1. Admin opens "Manage Users" and views the list of all accounts. 2. Admin searches for or selects a target user. 3. Admin chooses an action: view details, suspend, or delete. 4. System applies the action immediately. 5. If suspended or deleted, the target user's active sessions are invalidated and future login attempts are blocked. 6. Use case ends successfully.
Alternate Flow	3a. Delete Requires Confirmation At Step 3, if delete action selected: 3a1. Admin selects "Delete." 3a2. System prompts for explicit confirmation given the action is irreversible. 3a3. Admin confirms (resumes at Step 4) or cancels (process ends). 4a. Target Is the Only Remaining Admin At Step 4, if safety constraint triggered: 4a1. System detects the action would remove/suspend the last remaining Admin account. 4a2. System blocks the action to prevent the platform from being left without an administrator (process ends).

UC-17: View Platform Statistics | Actor: Admin

Goal	Inspect aggregate metrics and analytics across platform operations
Actor	Admin
Main flow	1. Admin opens the "Platform Statistics" dashboard. 2. System aggregates total events, total users, and total registrations from live data. 3. System displays the statistics, optionally broken down by date range or category. 4. Use case ends successfully.
Alternate Flow	2a. Aggregation Timeout Under Heavy Load At Step 2, if query processing times out: 2a1. Live aggregation exceeds an acceptable response time. 2a2. System falls back to a recently cached snapshot and indicates the data's "as of" timestamp (resumes at Step 3). 3a. No Data for Selected Range At Step 3, if range contains zero activity: 3a1. Admin selects a date range with no matching activity. 3a2. System displays zeros for that range rather than an error (resumes at Step 3).